

A Threshold-Aware Model for Early Sepsis Detection

Abstract. We report the first method achieving AUC 0.87 on a held-out cohort of 4821 patients, compared with AUC 0.79 for the standard early warning score. Sensitivity reached 87% at the operating threshold of 0.50, and the adjusted hazard ratio was HR 0.62 (95% CI 0.48 - 0.81).

$$h(t) = h_0(t) \exp(-0.48 x_1 + 0.15 x_2)$$

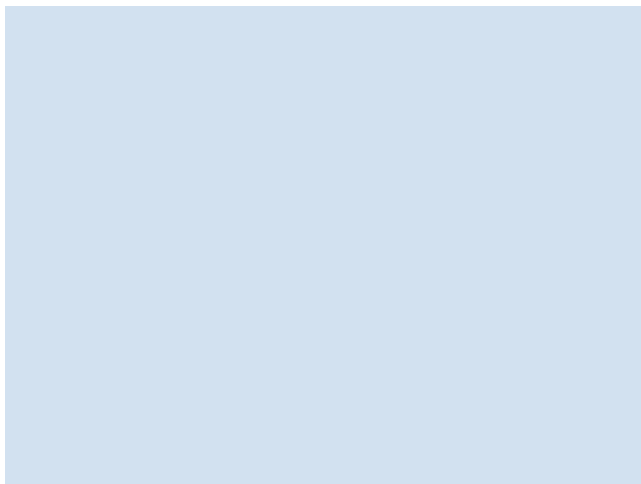


Figure 4: ROC curve of the proposed model.

The improvement over the baseline was consistent across all three validation sites, though the effect was attenuated in the smallest cohort where only 312 admissions were available for analysis.

3. Operating characteristics

Threshold	Sensitivity	Specificity
0.30	0.94	0.61
0.50	0.87	0.78
0.70	0.71	0.92

Table 2: Operating thresholds and their trade-off.

Lowering the threshold to 0.30 raises sensitivity to 94% but costs 17 points of specificity, which at a base rate of 6% implies roughly 3 false alarms for every true detection.